Dynamical Oceanography

Problem Set 2

1. Consider the steady nonlinear Stommel model in the form:

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = \nabla \times \tau - r\zeta \tag{1}$$

   Show that the nonlinear term will be small in the interior of the domain if $\beta L^2/U \gg 1$ where L is the scale of the large-scale motion and U a representative velocity. Show (e.g., by using using scale analysis) that for sufficiently strong wind, the nonlinear term must become more important. Derive a criterion (in terms of τ and the other parameters in the equation) for this.

2. Consider the Stommel model. Describe qualitatively how you might obtain nonlinear corrections to this, and what the nature of the corrections might be. Discuss the utility of this — how accurate the solutions might be, where they might fail, and so on.

   Optional: Using standard perturbation theory (e.g., by supposing that the solution is given by: $\psi = \psi_s + \psi'$ where ψ_s is the solution to the Stommel problem and ψ' is a small correction) explicitly obtain the nonlinear corrections to the Stommel solutions for a single gyre flow. Show that the effects of increasing nonlinearity are to move the center of the gyre poleward, and discuss the other properties of the solution. Where are the errors largest? [This problem is made much easier by using algebraic manipulation software, such as Maple.]

3. Beginning with the planetary geostrophic equations, or the shallow water equations, formulate a model equivalent to the Stommel model, but with $\beta = 0$. (i.e., the model is wind-driven, linear, vertically integrated, Rayleigh friction.) Obtain and discuss solutions. (You may need to do some reading on this.)

4. Use scaling theory on the planetary-geostrophic equations, or otherwise, to obtain an estimate for the depth of the wind-driven circulation in the ocean. Does this estimate differ if you use the quasi-geostrophic equations? Discuss.

5. Optional: Obtain solutions of the Stommel model in a circular or triangular domain.